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Honors 380: On Theory

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## Turning Phones into Campus IDs: Methodological Questions for a Next-Gen Attendance System

Every semester, I watch the same scene play out: someone makes it all the way to class, digs through their backpack, and suddenly realizes their student ID is missing. Sometimes they sprint back to their dorm, sometimes they just give up and hope the professor takes attendance manually. What rarely happens is that they forget their phone. That small observation is the seed of my senior project: a next-generation attendance and check-in system in which the phone serves as the ID. Instead of swiping plastic, students would tap their phone on an NFC reader at the door to be marked present or checked in.

On the surface, this sounds like a straightforward “better attendance sheet.” But as I dug deeper, I started to see it as a much more theoretical object: a small campus-level experiment in digital identity infrastructure, platform power, and everyday surveillance. Methodologically, that means my project cannot just be “write code until it works.” I need to define the field I am entering, limit the scope of what I am claiming to do, and make my design choices legible in terms of broader debates about convenience, privacy, and control. Honors 380 has made it hard to pretend this is just code; it is also a particular way of arranging people, devices, and institutions.

### **Defining the Field and the “Phone as ID” Problem**

The technical field around this project sits at the intersection of mobile computing, embedded systems, and security, plus a more applied domain of institutional identity systems. Chiang, Wang, and Chen’s article on an Android-based attendance system using GPS and NFC is the clearest example of how this space has already been formalized as a research area. Their system logs student presence via smartphone taps and location data, then evaluates reliability and user acceptance using quantitative metrics and surveys (Chiang, Wang, and Chen). That paper gives me a template for an “appropriate” methodology: build a working prototype, define what counts as success (error rate, latency, missed scans), and conduct a small evaluation with real students or instructors. My project follows that general pattern, but with a narrower focus. I am less interested in GPS-based geofencing and more in what happens when you try to ride on top of existing NFC ecosystems like Apple Pay and phone-based identity.

A key methodological step is to define what “phone as ID” actually means. There are at least three possibilities. First, I could try to treat payment cards the way my early Arduino experiments did: I used a PN532 reader and realized I could “see” Visa contactless cards at a low level, which felt like a shortcut to repurposing the tap signal as an attendance marker. EMVCo’s description of how EMV contactless chips generate cryptograms, follow specific “kernel” flows, and tightly couple to payment terminals makes it clear why this is a dead end for a serious system (EMVCo). Those cryptograms are designed to prevent exactly the kind of replay or re-identification that a naïve attendance system would need. Methodologically, I have to admit I am not building a gray-area EMV hack; I need a separate identity layer that coexists with, but does not piggyback on, payment rails.

The second option is the “wallet pass” paradigm on iOS. Apple’s documentation on `NFCVASReaderSession` outlines a sanctioned way to read non-payment passes—boarding

passes, loyalty cards, campus IDs—over NFC (Apple, “NFCVASReaderSession”). These VAS sessions are short-range, require special entitlements, and are tied to Wallet pass types. Apple’s security documentation on contactless passes explains how these passes can be configured to work independently or co-present with Apple Pay and how they are cryptographically protected (Apple, “Contactless Passes in Apple Pay”). Instead of seeing these constraints as purely technical, I read them as encoding a theory of identity: Apple wants ID-like interactions to flow through Wallet, which it mediates, rather than letting arbitrary apps treat payment cards as general-purpose NFC tokens. That means my “field” is not just NFC but the politics of closed mobile platforms. Methodologically, platform documentation becomes a primary source that defines what is even possible for a student developer.

### **The Design I Am Actually Building: NFC Stickers and Attendly**

The third option, and the one I am actually building now, is more platform-independent. Rather than expecting phones themselves to act as NFC tags, I use off-the-shelf NFC stickers (for example, NTAG215-style tags) mounted near classroom doors. The NFC Forum’s NDEF specification lays out how arbitrary payloads—URLs, opaque IDs, or app handoffs—can be encoded into NFC tags in an interoperable way (NFC Forum). In my current design, each classroom tag stores an NDEF URL that encodes “this is Class C” or “this is Session S of Class C.” When a student taps the sticker, their phone behaves as the NFC reader and opens that URL. From there, my second prototype, Attendly, takes over. Attendly is a web-based platform where students log in securely with their school, work, or partner-organization email; instructors create classes; and the system uses the combination of “this URL / this class session” plus “this logged-in account” to mark them present. In my original vision, the phone itself would be the cryptographic token—something closer to a campus card living inside Apple Wallet. In the

current iteration, the physical NFC tag acts more like a “check-in beacon” for the space. The phone contributes the tap gesture and the browser; identity is really managed in Attendly’s backend. That is less dramatic than a full wallet-pass architecture, but methodologically, it has two advantages. First, it works with both iPhones and Android devices using open NDEF patterns (NFC Forum). Second, it keeps the identity and policy logic in a space I can document—the web backend—rather than hiding it inside proprietary secure elements.

Real-world experience from practitioners is another kind of “text” I have to treat seriously. The Reddit thread “Opening front door with Apple Pay” captures what happens when hobbyists actually try to live in the “phone as key” future (Reddit). Users describe intermittent reads, confusion about when the wallet pops up, and the fragility of setups that depend on undocumented behavior. Methodologically, this pushes me to treat my own prototypes—both my Arduino-based NFC Attendance logger and my newer web-based Attendly backend—as systems that may feel smooth in controlled demos but behave messily in the wild. If the system misreads even a few times, people revert to old habits. That means in my evaluation, I cannot just measure averages; I have to pay attention to edge cases, cold starts, and the failure patterns that cause instructors to give up and go back to paper.

### **Power, Surveillance, and Data Minimization**

So far, I have mostly described the “how” of building and testing the system. But our class keeps insisting that we ask why this particular technical solution matters and what it does socially. At a theoretical level, phone-based attendance systems sit inside a familiar tension between frictionless convenience and deepening surveillance. If carrying one object—your phone, maybe with a tiny NFC sticker on the case—grants you access to classrooms, offices, and events, you gain efficiency, but you also make your daily routine easier to log, mine, and

misinterpret. Apple's security framing of contactless passes emphasizes short-range, user-initiated taps and strong cryptography (Apple, "Contactless Passes in Apple Pay"). That addresses the risk of eavesdropping or cloning a pass, but not the institutional question: who gets to see the logs of those taps, for how long, and for what purposes?

Here, Michel Foucault's account of modern institutions in *Discipline and Punish* becomes useful. Schools, prisons, and factories rely on detailed records of presence, absence, and performance to classify and normalize individuals (Foucault). An NFC-based attendance system simply makes those records cheaper to gather and easier to analyze. That does not automatically make the system evil, but it does mean that choices like how long I keep attendance logs, who can see full histories, and whether the data can be repurposed for behavior scoring or policing are central methodological decisions, not afterthoughts.

The technical literature frames my low-level methods section. Bleak's documentation explains asynchronous Bluetooth Low Energy patterns in Python for scenarios where a PN532 reader streams tag reads to a desktop or server process (Bleak Documentation), and Adafruit's PN532 guide gives me community-standard wiring and antenna advice (Adafruit). Together with Chiang, Wang, and Chen's evaluation methodology (Chiang, Wang, and Chen), these sources tell me how to wire, debounce, and measure the system. But methodologically, the more interesting question is what I let Attendly know and remember. For instance, I might decide that the system never stores precise timestamps finer than "class session," or that instructors can correct errors manually to avoid punishing students for hardware glitches. Those choices about logging and correction are where the theoretical concerns about discipline and fairness really show up in practice.

## **Rawls, Fairness, and Manual Overrides**

One way I keep the system from turning into pure tracking is by building friction back in on purpose. The most obvious example is the manual override: instructors will always have a back-end screen where they can correct attendance when a tag fails, the Wi-Fi is down, or a student was clearly in the room but never got through the login flow. The database is authoritative, but not absolute.

Here, John Rawls gives me a neat way to describe what I am trying to do. His idea of the “original position” behind a veil of ignorance asks us to imagine choosing rules for a system without knowing where we will land in it—whether we will be the most advantaged or the worst off (Rawls, “Original Position”). If I apply that thought experiment to Attendly, the question becomes: would I accept these rules if I might be the student with the newest iPhone and perfect connectivity, or the one with an old Android device, a dying battery, or spotty Wi-Fi? A Rawlsian mindset pushes me to treat things like manual override and alternative paths not as optional extras, but as fairness requirements. If I cannot tell a fair story about my design from behind the veil of ignorance—if there is no reasonable option for students who cannot or do not want to use a smartphone at all, whether that is a physical NFC card, a QR code, or a parallel sign-in process—then the method is not acceptable, no matter how slick the demo looks.

### **Agamben, Exceptions, and Students Who Fall Through the Cracks**

Anticipating challenges is one of the rubric’s criteria, and my sources make the roadblocks pretty clear. On the technical side, EMVCo makes it clear that I will never be able to treat a Visa card—or an Apple Pay token representing that card—as a stable identifier for attendance (EMVCo). Apple’s VAS documentation warns that production deployments require entitlements that small developers may not easily get (Apple, “NFCVASReaderSession”). The NFC Forum spec is dense and aimed at implementers, which means I will need to choose a

subset of NDEF features rather than pretending to implement the entire standard (NFC Forum). On the human side, the Reddit thread hints at user frustration when these systems half-work (Reddit). A professor who has to re-enter three or four names manually every class because a reader missed them may be more annoyed than if they had just passed around a paper sheet.

The hardest cases are the ones that fall between categories: the student whose tap is never recognized, whose account keeps getting misconfigured, or whose device simply does not cooperate. It would be easy, from a purely engineering perspective, to treat these as noise in the data. Giorgio Agamben's notion of the "state of exception" is a good reminder of why that would be a mistake. In *State of Exception*, he tracks how modern systems create zones where normal rules are suspended, and certain people become legally ambiguous—subject to power but not fully recognized by the law (Agamben). On a much smaller scale, a brittle attendance pipeline can accidentally produce micro "states of exception": students who are physically present but systematically left out of the record. If I quietly ignore them, I am effectively creating a group for whom the rule "show up and you are present" does not really apply. Methodologically, taking Agamben seriously means designing explicit exception-handling paths into Attendly: clear ways to mark "present but not recorded by the tap," visible logs of repeated failures so patterns cannot be hidden, and transparent rules about when and how human judgment overrides the machine.

## **Boundaries and Future Vision**

This project also forces me to think about what I am not doing. I am not designing a government-scale digital identity system, even though some of the technical ingredients (NFC, cryptography, wallet passes) overlap. I am also not promising a perfectly inclusive solution; a phone-based ID system obviously disadvantages people without modern smartphones or with inaccessible devices. Honors 380 keeps reminding us that every system has a boundary, and

those at the edges often feel the cost of design decisions the hardest. By explicitly defining my field as “campus and small-organization attendance,” relying on openly documented standards and platform rules, and building in spaces for human correction and exception, I am trying to design a methodology that keeps those boundaries visible rather than pretending they do not exist.

In that sense, this methodological essay is less about celebrating a clever NFC trick and more about building an honest roadmap for my senior project. The sources in my annotated bibliography function as different types of lenses: platform documentation that encodes hidden assumptions about identity and control, standards that define how data can move, hobbyist narratives that expose reliability issues, and research articles that model how to evaluate real deployments. Adding Foucault, Rawls, and Agamben gives me another set of lenses for thinking about power, fairness, and exceptions. My long-term hope is that Attendly becomes less a surveillance device and more a careful experiment toward a bigger idea—a secure, privacy-conscious app where people can keep all their IDs in one place for school, work, parking garages, and beyond, making everyday life a little easier for millions without forgetting what justice and limits require.



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